

I am a Canadian software engineer and ex-cofounder with 10 years of experience. Right now I lead building AI apps for mental health treatment in a large org. I am a fullstack engineer with significant experience in AI, ML, DevOps and even hardware programming. I can produce beautiful UIs and great products. I spend a lot of my free time working on side projects... recently I worked on Machine Learning in WebGPU, a Rust Solana High Frequency Trading system and my own programming language. Outside of that, I love working on my school bus camper project, woodworking, skiing and travelling with my fiancée.

Experience

OpenAI • Vercel AI SDK
• TypeScript • NextJS
• Azure • Terraform •
Tailwind
• Turborepo • Postgres
• Cosmos DB

EHN Canada • Software Engineering Team Lead

Oct 2024 - Now • Toronto (Remote)

I led a team of 7 engineers through various AI and devops projects. We developed a real-time voice AI + RAG Copilot for call center staff integrating with Salesforce/Genesys. I worked with medical experts to design a Speech to Speech + Video AI therapy app. We built a Mental Health Assessment platform and a variety of integrations between systems like Snowflake, Salesforce, Genesys and Kipu EMR. I implemented comprehensive CI/CD + DevOps systems with tools like Terraform CDKTF, Azure and Sentry. I designed our event driven architecture involving websockets, audio, streaming AI and serverless.

OpenAI • Vercel AI SDK
• TypeScript • NextJS
• Vercel • PostGRES
• Sentry • Tailwind

VentureKit • Senior Software Engineer

May 2024 - Oct 2024 • Toronto (Remote)

I built a variety of streaming AI powered features in Vercel, Next, Typescript, Tailwind and AI SDK. These include an AI + web scraping competitor search tool, AI RAG market research reports with graphs, and AI generated Guides. I also migrated our stack from Next 13 to Next 15.1 and fixed a variety of bugs, supporting 300,000 users.

Vercel AI SDK • TypeScript
• OpenAI • NextJS • Vercel

ALR CPA • Software Engineer / Head of Tech

July 2023 - Oct 2024 • Toronto (Remote)

Built and launched customer success web platform. Updated hiring process, policies and remote employee support, making 4 successful new hires. Improved SEO, content marketing and launched new analytics, drastically improving web vitals/lighthouse scores. Improved SOPs and infrastructure towards SOC 2 audit. Implemented CRM, sales process and content marketing strategy with AI. trained team on AI tools.

Anytype • Senior Software Engineer

October 2022 - May 2023 • Berlin (Remote)

TypeScript • React •
Electron • Go • IPFS •
CRDT • gRPC

P2P notion-like platform with >200k users and >4k stars on github. Helped push through to public launch, developing the desktop app and browser extension. Lead developer for the widgets and onboarding systems. Supported end users on the forum, fixed bugs and promoted code quality improvements.

Qualio • Full-Stack/Devops Engineer

September 2020 - August 2021 • San Francisco (Remote)

QA platform for Life Sciences/Pharma. Developed customer migration and onboarding system, eliminating manual labor required from the engineering team. Transitioned spaghetti monolith to our microservice architecture, implementing a deployment pipeline to achieve high availability and scaling OKRs.

Python • Typescript • React
• Node • AWS • Lambda •
Docker • CloudFormation

Huckleberry AI • Co-founder/CTO (seed stage)

January 2019 - September 2020 • Waterloo (Hybrid)

Raised ~150k, taking our 2 products from idea to paying customers. Products include ping.buzz (b2b SaaS) + huckleberry.app and web extension (b2c). Managed a small dev team + 12 co-op students from Waterloo. Designed our serverless, hexagonal, functional DDD architecture and built our data pipeline to process market feed data.

TypeScript • Node • React
• Native • PostGres • Redis
• AWS

RCO Ventures • Software Developer

April 2017 - September 2020 • Waterloo (Remote)

Started a contracting firm with over 30+ startup clients. Projects included web/mobile applications, data processing, devops and digital transformation. Projects include AI powered healthcare billing app, time tracking software, a home services app, bid management platform, gig economy platform, data-analysis for a class-action lawsuit and aggregated datasets as a service.

Python • TypeScript •
Angular • React • Golang •
C# • Netlify • Vercel • AWS
• Mongo • PostGres • Ruby
• RabbitMQ • NextJS •
Kubernetes • GraphQL

Emrys Analytics • Director of Analytics

September 2016 - March 2017 • Waterloo (Hybrid)

Developed risk models and underwriting software to insure healthcare facilities against cyber risk in the US. Built an underwriting engine written in Rust/Python. Built an ETL pipeline to aggregate and clean datasets. Applied models such as k-means clustering and monte carlo. Co-authored paper on autonomous vehicle insurance data for Ford.

Python • TypeScript • Rust
• Electron • Postgres •
AWS

Diebold Nixdorf • Software Architecture R&D Internship

May 2016 - August 2016 • London

Developed redis session authentication middleware for ATMs/Banking Machines. Built proof of concepts and conducted experiments to support decision-making with webRTC, websockets and http/2.

C# • TypeScript • Angular •
Redis

DELTAfi (Δfi) • Founder/Hardware Designer

May 2013 - May 2016 • London

Designed and built various synthesizers and other embedded projects from PCB prototyping to small run production.

C • KiCAD • Python

Education

Western University / Comp. Sci. and Mathematics B.Sc.

2014 - 2016, Canada